

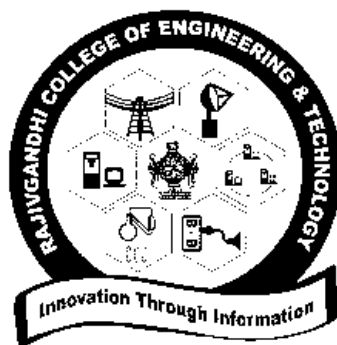
RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY

KIRUMAMPAKKAM – PUDUCHERRY - 607403

DEPARTMENT OF CHEMISTRY

CHEMISTRY MANUAL

(First year B. Tech)



NAME :

ROLL NO :

BATCH :

SECTION :

DEPT :

PONDICHERRY UNIVERSITY

CHEMISTRY LAB – Syllabus

1. Estimation of Iron by Colorimetry.
2. Percentage composition of sugar solution by Viscometry
3. Determination of Lead in polluted water by Conductometry.
4. Determination of Chemical Oxygen Demand of water sample.
5. Determination of Dissolved Oxygen in water.
6. Determination of total hardness of water by EDTA method.
7. Estimation of Magnesium by EDTA.
8. Estimation of ferrous by Permanganometry.
9. Estimation of available Chlorine in Bleaching powder.
10. Determination of Copper in copper sulphate solution.
11. Estimation of mixture of alkali
12. Estimation of acetic acid.
- ~~13. Estimation of chlorides in water.~~

LIST OF EXPERIMENTS

I NEUTRALISATION TITRATION

ACID – BASE TITRATION

1. Estimation of mixture of alkali
2. Estimation of percentage of acetic acid in vinegar

II REDOX TITRATION

PERMANGANOMETRY

3. Estimation of ferrous by permanganometry

IODOMETRY

4. Determination of Dissolved Oxygen in water
5. Estimation of available Chlorine in Bleaching powder
6. Determination of Copper in Brass

III PRECIPITATION TITRATION

ARGENTOMETRY

7. Estimation of Chlorides in water by Mohr's method

IV COMPLEXOMETRIC TITRATION

EDTA Titration

8. Determination of total hardness of water by EDTA method
9. Estimation of Magnesium by EDTA

V DEMONSTRATION EXPERIMENTS

1. Percentage composition of sugar solution by Viscometry
2. Determination of Lead in polluted water by Conductometry
3. Determination Chemical Oxygen Demand

GUIDELINES TO BE FOLLOWED IN CHEMISTRY LABORATORY

The following guidelines should be adopted to avoid most of the accidents occurring in the Chemistry laboratory.

1. Students should listen carefully the instructions given in the practical classes.
2. Laboratory coat must be worn at all times during the laboratory period.
3. Do not perform unauthorized experiments.
4. Record your observations as and when you proceed.
5. Many laboratory chemicals are poisonous. Hence the students are advised not to taste any chemical or solution.
6. Read the label carefully before taking a chemical from its container.
7. Replace the reagent bottles in their respective places after use.
8. Never pour the reagents back into the respective reagent bottle.
9. When diluting concentrated acids, pour the acid slowly into the water with constant stirring Never add water to the acid.
10. Sink is only for draining liquid. Do not throw any solid material into the sink.
11. Do not use cracked or broken glassware.
12. Use tongs or towels to handle hot objects.
13. Spatula should be used to handle chemicals. Never use your fingers.
14. Any accident or breakage during the experimental work should be reported to the Faculty In charge or Lab Instructor immediately.
15. Keep gas and water taps closed when not in use.
16. After completing the experiment, wash every apparatus, disconnect all electrical connections, keep the apparatus in proper places and clean your work table before leaving the laboratory.
17. Submit Observation note book and Record note book within a week time.

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Expt. No.	Date	Name of the Experiment	Remarks	Signature
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FUNDAMENTALS OF VOLUMETRIC ANALYSIS

Two major methods are available in the analysis of substances are

Qualitative Analysis

It deals with the analysis of the quality or chemical nature of the substance by determining the acidic and basic radicals.

Quantitative Analysis

It enables us to find out the quantity or exact amount of a substance present in a given sample.

Quantitative analysis is classified into two types

Gravimetric Analysis

In this, the quantity of the substance is determined by methods, which make use of direct weighing.

Volumetric Analysis

It is the method of finding out the quantity of the substance from the measurement of volumes of reacting solution.

In volumetric analysis, the concentration of a solution is determined by using another suitable solution whose concentration is known accurately.

Terms used in Volumetric Analysis

Titration

The process of determining the strength of an unknown solution (taken in the conical flask) by allowing it to react with a standard solution (solution of known strength taken in the burette) is known as Titration. The completion of the reaction is monitored by the use of an indicator.

Titrant

The solution containing a known weight (known concentration) of a substance is called a Titrant.

Titrate

The solution which contains a substance to be estimated (unknown concentration) is known as a Titrate.

Standard Solution

When the exact amount of the substance present in a definite volume of the solution is known, it is called a standard solution.

(i) Primary standard: The substance which is unaltered in air during weighing is known as Primary standard substance. Solutions of known concentration, by weighing a definite amount of primary standard substances are called primary standard solution. These primary substances should not be hygroscopic nor oxidized by air.

Ex: Na_2CO_3 , $\text{K}_2\text{Cr}_2\text{O}_7$ and NaCl .

(ii) Secondary standard: The substance whose composition is altered is called as Secondary standard substance. A solution of approximate concentration which varies with time is called Secondary solution and its contents of the active substance will be found by comparison against a primary standard.

Ex: NaOH and KMnO_4 .

Concentration of a solution is expressed in various ways.

(a) Normality (N): The normality of a solution is the number of gram equivalent of the solute per litre of the solution.

Normality (N) = No. of gram equivalent of solute / Volume of solution (in 1000 ml)

(b) Molarity (M): The molarity of a solution is the number of moles of the solute per litre of the solution.

Molarity (M) = No. or moles of solute / Volume of solution (in 1000 ml)

(c) Molality (m): The molality of a solution is the number of moles of the solute per 1000 g of the solvent.

Molality (m) = No. of moles of solute / Mass of solvent in Kg

Indicator

It is a substance which when added to titrate, indicates the completion of reaction by changing its colour at the end point.

End Point

This is the point in the titration, which indicates the completion of a chemical reaction.

Standardisation

It means determination of the strength of an unknown solution with the help of a standard solution.

Estimation

To find out the amount of a substance present in a certain volume of the given solution.

Solute

The substance, which dissolves into the solution, is called as a solute.

Normal solution (1N solution)

A normal solution is a solution containing one gram equivalent weight of the solute in one litre of the solvent.

Deci-normal solution (N/10 solution)

A solution containing one tenth of a (1/10) gram equivalent of the solute in one litre of the solution is known as deci-normal solution.

Principle involved in volumetric analysis

Volumetric analysis is based on the law of volumetric analysis. When two solutions completely react with each other, the product of volume and normality of one solution will be equal to the product of volume and normality of the other solution.

$$V_1N_1=V_2N_2$$

V_1 – Volume of the first solution

N_1 – Normality of the first solution

V_2 – Volume of the second solution

N_2 – Normality of the second solution

Important Calculations

Normality (Strength) of a solution =

$$(\text{Weight of the solute in one litre}) / (\text{Equivalent weight of the solute})$$

Weight of the solute in one litre =

$$(\text{Equivalent weight of the solute}) \times (\text{Normality of the solution})$$

Weight of solute present in y ml of the solution =

$$\text{Equivalent weight of solute} \times \text{Normality of the solution} \times (y/1000)$$

Types of the reaction involved in volumetric analysis

1. Neutralization reaction
2. Oxidation – Reduction reactions
3. Precipitation titrations
4. Complexometric titrations

1. Neutralisation reactions

The reaction between an acid and a base resulting in the formation of salt and water is known as neutralization reaction.

The acid is titrated against the base (or vice versa) in the presence of an indicator. The indicator will have one colour in acid medium and entirely different

colour in basic medium i.e., the colour changes with changes in pH of the solution.

Two types of titrations are

Acidimetry: Estimation of amount of acid

Alkalimetry: Estimation of amount of an alkali

Common indicators used are methyl orange, Phenolphthalein, methyl red etc.

2. Oxidation – reduction reactions

They are also called redox reactions i.e., the process of oxidation and reduction takes place simultaneously. The commonly used oxidizing agents are Potassium permanganate, Potassium dichromate and the reducing agents are oxalic acid, ferrous ammonium sulphate etc. These reactions are classified into three types.

(a) Permanganametry

Estimation of the substances by using permanganate as an oxidizing agent

(b) Dichrometry

Estimation of the substances by using dichromate as an oxidizing agent

(c) Iodometry

Estimation of substances with reducing properties by using liberated iodine.

3. Precipitation titration

In the precipitation titration, one of the reacting constituent is separated in the form of a precipitate during the course of titration.

4. Complexometric titration (EDTA titration)

Estimation of the solution involving complex formation by using a complexing agent EDTA and a complexing indicator Eriochrome Black-T.

Apparatus used in Volumetric Analysis

Burette, Pipette, Conical flask, Volumetric flask, Funnel, Measuring Jar, Weighing

Bottle

Abbreviations

V	–	Volume
ml	–	millilitre
gm	–	grams
N	–	Normality
Conc	–	Concentrated

dil – diluted
ppm – parts per million

Equivalent weights of some Inorganic substances

Substance	Eq.wt.
NaHCO ₃	84
Na ₂ CO ₃	53
K ₂ Cr ₂ O ₇	49.04
FeSO ₄ . 7 H ₂ O	278
FeSO ₄ .(NH ₄) ₂ SO ₄ . 6 H ₂ O	392.2
Na ₂ S ₂ O ₃ . 5 H ₂ O	248.2
CuSO ₄ . 5 H ₂ O	249.7
AgNO ₃	170
NaCl	58.44
CaCO ₃	50
Ferrous iron	55.85
Copper ion	63.54

Strength of Common acids

Acid	Strength (N)
HCl	12
HNO ₃	16
H ₂ SO ₄	36
Acetic acid	17
Ammonia	14

Indicators and their pH range

Indicator	pH range	Acidic	Basic
Methyl Orange	3.1 – 4.4	Red	Orange
Phenolphthalein	8.3 – 10	Colourless	Red

Viva Question and Answer

1. Define Qualitative Analysis

It deals with the analysis of the quality or chemical nature of the substance by determining the acidic and basic radicals.

2. Define Quantitative Analysis. Classify the Quantitative Analysis.

It enables us to find out the quantity or exact amount of a substance present in a given sample. Quantitative analysis is classified into two types

(i) Gravimetric analysis

(ii) Volumetric analysis

3. What is the Titrate?

The solution which contains a substance to be estimated (unknown concentration) is known as a Titrate.

4. What is Titrant?

The solution containing a known weight (known concentration) of a substance is called a Titrant.

5. What is Titration?

The process of determining the strength of an unknown solution (taken in the conical flask) by allowing it to react with a standard solution (solution of known strength taken in the burette) is known as Titration.

6. What is Standard Solution?

When the exact amount of the substance present in a definite volume of the solution is known, it is called a standard solution.

7. What are the Classification of Standard Solution?

(i) Primary standard: The substance which is unaltered in air during weighing is known as Primary standard substance

Eg : Oxalic Acid, Na_2CO_3 , $\text{K}_2\text{Cr}_2\text{O}_7$ and NaCl .

(ii) Secondary standard: The substance whose composition is altered is called as Secondary standard substance. A solution of approximate concentration which varies with time is called Secondary solution.

Eg : NaOH and KMnO_4 , $\text{Na}_2\text{S}_2\text{O}_3$

8. Define Normality.

The normality of a solution is the number of gram equivalent of the solute per litre of the solution.

Normality (N) = No. of gram equivalent of solute / Volume of solution (in 1000 ml)

9. Define Molarity.

The molarity of a solution is the number of moles of the solute per litre of the solution.

Molarity (M) = No. or moles of solute / Volume of solution (in 1000 ml)

10. Define Molality.

The molality of a solution is the number of moles of solute per 1000 g of the solvent.

Molality (m) = No. of moles of solute / Mass of solvent in Kg

11. Define Indicator.

The substance which when added to the titrate, indicates the completion of reaction by changing its colour at the end point is known as Indicator.

12. Define End Point.

The point at which the reaction completes and the completion of the reaction can be identified by the colour change with the help of indicator

13. What is meant by Standardisation?

It means determination of the strength of an unknown solution with the help of a standard solution.

14. Define Solute.

The substance, which dissolves into the solution, is called as a solute.

15. What are the types of reaction involved in Volumetric Analysis?

- | | |
|-----------------------------|------------------------------------|
| 1. Neutralization reaction | 2. Oxidation – Reduction reactions |
| 3. Precipitation titrations | 4. Complexometric titrations |

16. What is meant by Neutralization Reactions?

The reaction between an acid and a base resulting in the formation of salt and water is known as neutralization reaction. Two types of titrations are,

Acidimetry: Estimation of amount of acid

Alkalimetry: Estimation of amount of an alkali

Common indicators used are methyl orange, Phenolphthalein, methyl red etc.

17. Classify the Oxidation – Reduction Reactions.

These reactions are classified into three types.

- (a) Permanganometry: Estimation of substances by using permanganate as an oxidising agent

- (b) Dichrometry: Estimation of substances by using dichromate as an oxidising agent
- (c) Iodometry: Estimation of substances with reducing properties by using liberated iodine.

21. What is meant by Precipitation titration?

In the precipitation titration, one of the reacting constituent is separated in the form of a precipitate during the course of titration.

22. What is meant by pH.

pH = Power of hydrogen (or) it is the negative log of concentration of hydrogen ions

$$\text{pH} = -\log [\text{H}^+]$$

23. Indicators and their pH range.

Indicators	pH range
1. Methyl Orange	3.1-4.4
2. Phenolphthalein	8.4-10
3 EBT	7.0 -11

24. Types of titration used in indicators.

- 1. Acid – Base Titrations - Phenolphthalein, Methyl Orange
- 2. Iodometry Titrations - Starch
- 3. Complexometric Titrations - EBT (Eriochrome Black-T)

25. What is the Principle involved in volumetric analysis?

Volumetric analysis is based on the law of volumetric analysis. When two solutions completely react with each other, the product of volume and normality of one solution will be equal to the product of volume and normality of the other solution.

$$V_1 N_1 = V_2 N_2 \quad V_1 - \text{Volume of the first solution}$$

N_1 – Normality of the first solution

V_2 – Volume of the second solution

N_2 – Normality of the second solution

Table I

Sl. No.	Concentration of iron solution	Optical Density
	Unknown	

Expt. No:

Date:

ESTIMATION OF IRON BY COLORIMETRY

Aim

To determine the amount of iron present in the given solution: Colorimetrically.

Materials Required

Colorimeter

Standard ferrous solution

Potassium thiocyanate solution

dil. HNO_3

Principle

Calorimetric method is convenient for systems in which substances or their solutions are coloured. The intensity of colour can easily be measured by using a Photoelectric Calorimeter. When a species under test is colourless, than a suitable complexing agent is added to the solution so that a coloured complex which can absorb light in the visible region is obtained.

Iron when complexed with thiocyanate gives an intense red color. The intensity of the colour depends upon the concentration of iron in the solution.

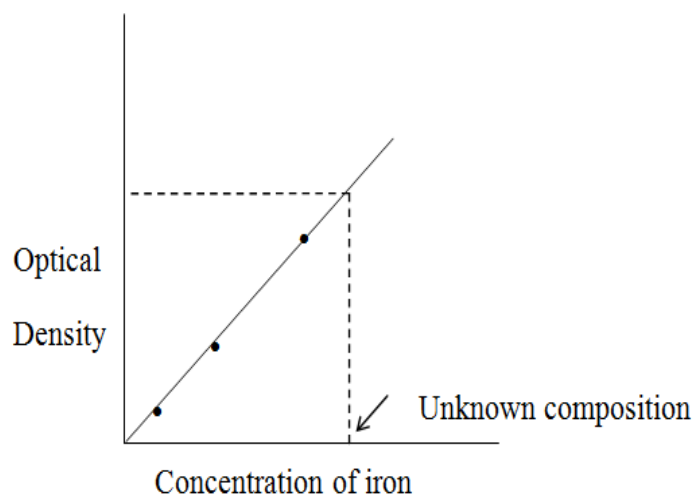
A calibration curve is obtained by plotting optical density versus concentration. The amount of iron in the unknown sample is determined from the plot using observed optical density.

Procedure

Optical density or Absorbance is measured at 490 nm by selecting a proper filter. The calorimeter is set at zero absorbance using a blank solution, which contains all reagents except metal ions. Different concentrations of the standard iron solution say 0.05, 0.1, 0.15, 0.3 M are prepared in 100 ml of standard flasks. 5 ml of potassium thiocyanate solution is added followed by 2 ml of 4N nitric acid to each of the iron solution samples.

The mixture in the standard flask is shaken well and a portion of the coloured solution is taken in to the cell. The optical density of each standard solution is measured using the same filter.

Model Graph



Calculation of amount of Iron

Amount of Ferrous

$$= \text{Concentration of ferrous} \times \text{Eq. Wt. of ferrous (55.85)} \times (100/1000)$$

$$= \text{_____ N} \times 55.85 \times (100/1000)$$

Amount of ferrous = _____ g

The optical density is plotted against concentration. This plot is called calibration curve and will be a straight line passing through origin as per the Beer-Lamberts law.

Now the optical density of the solution under test is similarly measured using the same cell and same filter. From the calibration curve, the concentration of unknown iron solution and hence the amount of iron can be calculated.

Result

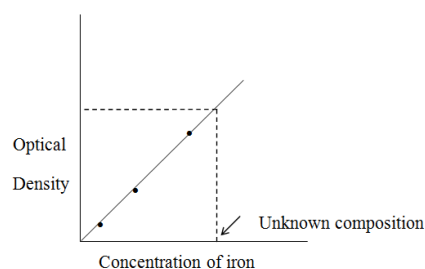
The amount of iron present in the given sample = _____ g

Viva

1. Which complexing agent is used in the estimation of iron by colorimetry?

Potassium thiocyanate solution is used as the complexing agent. Iron when complexed with thiocyanate gives an intense red color. The intensity of the colour depends upon the concentration of iron in the solution.

2. Draw the model graph in the estimation of iron by colorimetry.



3. Mention the wavelength of iron.

490 nm

4. Define Beer Lambert's law.

The intensity of a beam of monochromatic light decreases exponentially as the concentration of the absorbing substance in solution and the thickness increases independently.

$$A = \log (I_0 / I_t) = \epsilon^{ct}$$

Table I[illegible]

Expt. No:

Date:

ESTIMATION OF PERCENTAGE COMPOSITION OF SUGAR SOLUTION BY VISCOMETRY

Aim

To estimate the percentage composition of sugar solution using Ostwald Viscometer.

Materials Required

1. Ostwald Viscometer
2. Stop clock

Principle

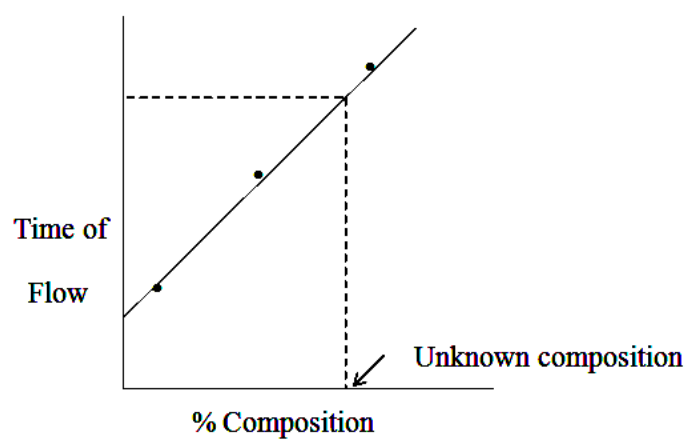
By plotting the values of % composition against time of flow in sec we get a curve from which unknown solution composition is determined. Also by plotting the values of viscosities of solutions against their concentration we get a curve from which the concentration of unknown solution is determined.

Procedure

The viscometer is first thoroughly cleaned and dried. A definite quantity of water is introduced into the bulb B and sucked up through the capillary into the smaller bulb. Now bring water to touch the mark in bulb A and hold it by placing finger at the top of the narrow limb. Remove the finger and start stopwatch. Stop it as soon as water touches the mark. Repeat this process 2 – 3 times and take the mean value.

Prepare a series of solutions of sugar (1%, 2%, 3%, ... 8%) using distilled water. The time of flow of each solution is noted by using stopwatch. The time of flow of unknown solution is also measured.

Model Graph



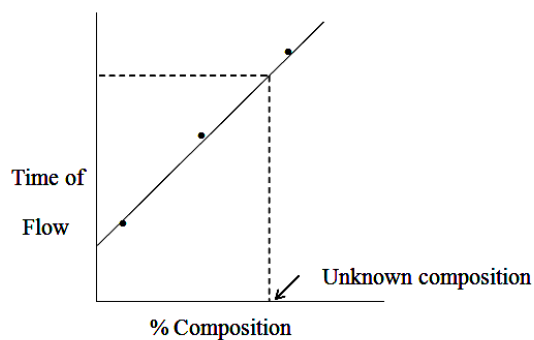
A curve is plotted between the % composition and time of flow in sec. A straight line is obtained. The composition of unknown solution is calculated by locating and marking the point on the straight line corresponding to the measured time of flow. A perpendicular is drawn from that point on the % composition axis from which composition of unknown solution can be read directly.

Result

The percentage composition of unknown sugar solution = _____ %

Viva

1. Draw the model graph in the estimation of iron by colorimetry.



Expt. No:

Date:

DETERMINATION OF LEAD BY CONDUCTOMETRY

Aim

To determine the amount of lead present in the given sample by conductometric method.

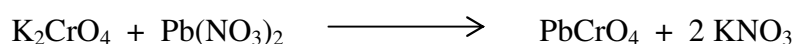
Materials Required

1. Conductometer
2. Conductivity Cell
3. Standard potassium chromate solution
4. Lead nitrate solution

Principle

Solution of electrolytes conducts electricity due to the presence of ions. The specific conductance of a solution is proportional to the concentration of ions in it.

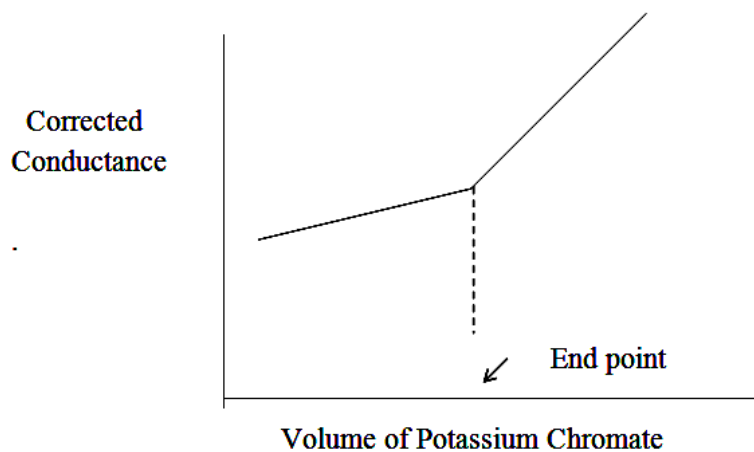
When lead nitrate solution is titrated with potassium chromate the conductance increases slightly. It is due to the replacement of nitrate ion by chromate ion.



After the end point the conductance increases rapidly due to the further addition of potassium chromate.

A graph is drawn between volume of potassium chromate added and the conductance of the solution. The end point is the intersection of the two lines

Model Graph



Calculation of amount of Lead

Volume of Potassium chromate (V_1) = _____ ml (from graph)

Normality of Potassium chromate (N_1) = _____ N

Volume of Lead solution (V_2) = 20 ml

Normality of Lead (N_2) = ?

$$V_1 N_1 = V_2 N_2 \quad N_2 = V_1 N_1 / V_2$$

Normality of Lead (N_2) = { Volume of Potassium chromate x Normality of Potassium chromate } / Volume of Lead solution

$$N_2 =$$

Normality of Lead (N_2) = _____ N

Amount of Lead in whole of the given solution

$$= \text{Normality of Lead} \times \text{Eq. wt. of lead (103.4)} \times (100/1000)$$

=

=

Amount of Lead present in the given solution = _____ g

Procedure

The given lead nitrate solution is made up to the mark in a 100 ml volumetric flask. 20 ml of the lead solution is taken in a 100 ml beaker and it is diluted by adding 20 ml of conductivity water and conductivity cell is immersed in the solution. Conductance is measured periodically after each addition of potassium chromate solution. Conductance values are recorded and the observed conductance is corrected for dilution effect.

The value of corrected conductance is plotted against the volume of titrant added. From the graph the end point is determined and hence the amount of lead present in the given solution is calculated.

Result

Amount of lead present in whole of the given solution = _____ g

Titration I**Standardization of FAS Solutoin**

Burette Solution	-	FAS Solution
Pipette Solution	-	10 ml of standard $K_2Cr_2O_7$
Additional solution	-	10 ml distilled water, 10 ml of H_2SO_4
Indicator	-	2 to 3 drops of Ferroin
End point	-	Organic blue to reddish brown colour

Sl. No.	Volume of $K_2Cr_2O_7$ (V_1) (ml)	Burette Reading (ml)		Volume of FAS (V_2) (ml)	Concordant titre value (ml)
		Initial	Final		
1	10				
2	10				
3	10				

Volume of $K_2Cr_2O_7$ $V_1 = 10$ ml

Normality of $K_2Cr_2O_7$ $N_1 =$ N

Volume of FAS $V_2 =$ ml (titre value)

Normality of FAS $N_2 = ?$

$$V_1 N_1 = V_2 N_2$$

$$N_2 = V_1 N_1 / V_2$$

$$\text{Volume of } K_2Cr_2O_7 \times \text{Normality of } K_2Cr_2O_7$$

$$\text{Normality of FAS (N}_2\text{)} = \frac{\text{Volume of } K_2Cr_2O_7 \times \text{Normality of } K_2Cr_2O_7}{\text{Volume of FAS}}$$

$$N_2 =$$

$$\text{Normality of FAS (N}_2\text{)} = \text{N}$$

Expt. No:

Date:

DETERMINATION OF CHEMICAL OXYGEN DEMAND

Aim

To determine the chemical oxygen demand value of a given water sample by volumetric method.

Chemicals Required

1. Standard $K_2Cr_2O_7$
2. Conc. H_2SO_4
3. Standard FAS
4. Ferroin indicator

Principle

Most types of organic matter are oxidized by a boiling mixture of chromic and sulfuric acids. A sample is refluxed in strongly acid solution with a known excess of potassium dichromate. After digestion the remaining unreduced $K_2Cr_2O_7$ is titrated with ferrous ammonium sulphate (FAS). The amount of $K_2Cr_2O_7$ consumed for the oxidation of organic matter is determined and the amount of oxidisable organic matter is calculated in terms of oxygen equivalent.

Procedure

Titration I Standardization of FAS

A standard solution of $K_2Cr_2O_7$ is prepared in a 100 ml std.flask. 10 ml std $K_2Cr_2O_7$ solution is diluted to 100 ml of conc. H_2SO_4 is added. After cooling, the solution is titrated with ferrous ammonium sulphate taken in the burette with the addition of 2 to 3 drops of ferroin indicator. The titration is repeated for the concordant values.

Titration II**Determination of COD**

Burette Solution	-	FAS Solution
Pipette Solution	-	20 ml of Water Sample
Additional solution	-	10 ml of $K_2Cr_2O_7$, 10 ml of H_2SO_4
Indicator	-	2 to 3 drops of Ferroin
End point	-	Organic blue to reddish brown colour

Sl. No.	Volume of Water (V_1) (ml)	Burette Reading (ml)		Volume of FAS (V_2) (ml)	Concordant titre value (ml)
		Initial	Final		
1	Sample (A)				
2	Sample (B)				

$$\text{COD of given sample water} = \frac{\text{Normality of FAS} \times (A-B) \times 8 \times 1000}{\text{Volume of Sample}}$$

=

$$\text{COD of given sample water} = \quad \text{ppm}$$

Titration II**Determination of COD**

20 ml of sample water and 20 ml of distilled water are taken in two separate round bottom flask. The former is the sample and later is the blank. To both of them 10 ml of $K_2Cr_2O_7$ and 10 ml of concentrated H_2SO_4 are added and refluxed for 2 hrs. After refluxion they are allowed to cool for about half an hour and are titrated against std.FAS using ferroin indicator. The first sharp color change from bluish green to reddish brown is taken as the end point of the titration. From the difference between the two-titre values COD of water sample is calculated.

Result

The chemical oxygen demand of given water sample is = _____ ppm

Viva**1. Define COD.**

The amount of oxygen required by organic matter in a sample of water for its oxidation by a strong chemical oxidizing agent such as $K_2Cr_2O_7$. It is expressed as ppm.

2. Write the formula used to calculate COD.

$$\text{COD of given sample water} = \frac{\text{Normality of FAS} \times (A-B) \times 8 \times 1000}{\text{Volume of Sample}}$$

3. What is the indicator used in the COD experiment?

Ferroin is the indicator.

Titration I**Standardization of EDTA**

Burette Solution	-	EDTA
Pipette Solution	-	20 ml of Standard hard water
Indicator	-	Eriochrome Black– T
End point	-	Colour change from wine red to blue
Additional Solution	-	2 ml of ammonia buffer solution

Standard Hard Water Vs EDTA

Sl. No.	Volume of Standard hard water (ml)	Burette Reading (ml)		Volume of EDTA (V_1) (ml)	Concordant titre value (ml)
		Initial	Final		
1	20				
2	20				
3	20				
4	20				

Volume of Standard hard water = 20 ml

Volume of EDTA solution consumed (V_1) = _____ ml (titre value)

1 ml of standard hard water contains 1 mg of calcium carbonate

20 ml of standard hard water contains 20 mg of CaCO_3

20 ml of standard hard water consumes V_1 ml of EDTA

V_1 ml of EDTA solution = 20 mg of CaCO_3

1 ml of EDTA solution = $(20 / V_1)$ mg of CaCO_3 equivalent

= $(20 / \text{_____})$ ml of CaCO_3 equivalent

= _____ mg of CaCO_3

Expt. No:

Date:

ESTIMATION OF TOTAL HARDNESS OF WATER BY EDTA METHOD

Aim

To estimate the total hardness in the given sample of hard water by EDTA method.

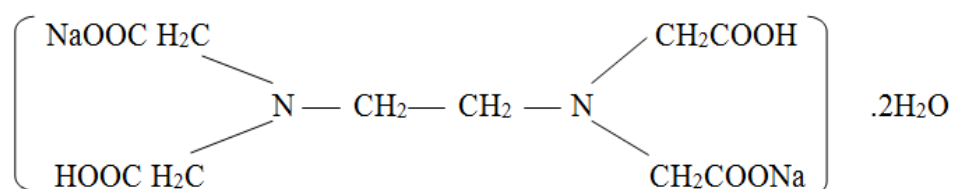
Chemicals Required

Standard hard water, EDTA, Eriochrome black – T indicator, Ammonia buffer solution

Principle

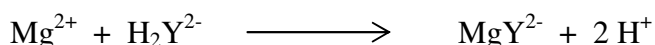
The hardness of water is due to dissolved bicarbonates, carbonates (Temporary hardness) and chlorides, sulphates of calcium and magnesium (Permanent hardness).

Ethylene Diamine Tetra Acetic acid (EDTA) is a tetra carboxylic acid. This method is based upon the fact that EDTA forms remarkable stable complexes with metals, particularly di and polyvalent metals and it can be represented as,

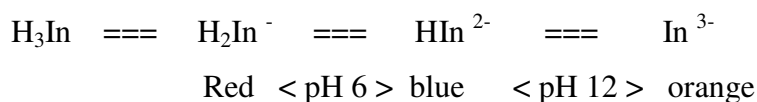


EDTA does not dissolve completely in water and hence it's disodium salt is used to estimate the various hardness producing ions present in the hard water. EDTA is abbreviated as $\text{Na}_2\text{H}_2\text{Y}$ and its dissociation at pH 10 is, $\text{Na}_2\text{H}_2\text{Y} \longrightarrow \text{H}_2\text{Y}^{2-} + 2 \text{Na}^+$

EDTA reacts with Ca and Mg ions forming stable complexes.



The indicator Eriochrome Black – T is a tribasic acid dye represented by H_3In which dissociates to different coloured ions at various pH.



The bivalent blue coloured indicator ion which exists at about pH 10 reacts with Mg^{2+} , producing red coloured Mg – indicator complex. $\text{Ca} - \text{EDTA complex} > \text{Mg} - \text{EDTA complex} > \text{Mg} - \text{indicator complex}$.

Titration II**Estimation of Total Hardness of Water sample**

Burette Solution	-	EDTA
Pipette Solution	-	20 ml of sample hard water
Indicator	-	Eriochrome Black– T
End point	-	Colour change from wine red to blue
Additional Solution	-	2ml of ammonia buffer solution

EDTA Vs Sample Hard Water

Sl. No.	Volume of Water sample (V ₁) (ml)	Burette Reading (ml)		Volume of EDTA (V ₂) (ml)	Concordant titre value (ml)
		Initial	Final		
1	20				
2	20				
3	20				
4	20				

Volume of hard water sample (V₁) = 20 ml

Volume of EDTA consumed (V₂) = _____ ml (titre value)

20 ml of the given hard water sample consumes V₂ ml of EDTA

20 ml of the given hard water sample contains $(20 / V_1) \times V_2$ mg of CaCO₃

1000 ml of given hard water sample contains

$$= (20 / V_1) \times V_2 \times (1000 / 20) \text{ mg of CaCO}_3$$

$$= 1000 \times (V_2 / V_1) \text{ mg of CaCO}_3$$

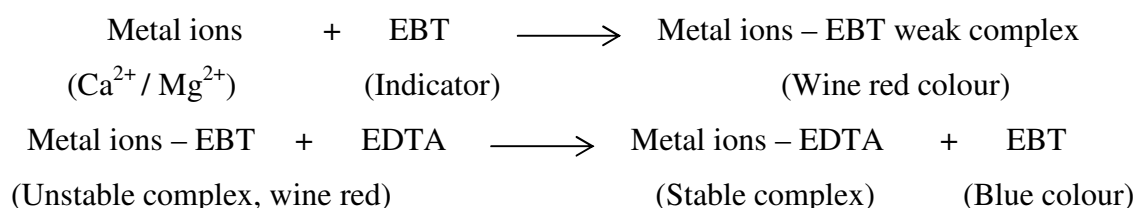
ml

$$= 1000 \times \frac{\quad}{\quad}$$

20 ml

Total hardness of the given sample of hard water = _____ ppm

So if a drop of indicator is added to hard water sample containing Ca and Mg ions at pH 10, the Mg – indicator complex will be formed producing red colour. Now to this solution, if EDTA solution is gradually added it will remove Ca – ions first because Ca – EDTA complex being more stable. When all calcium ions are used up, excess EDTA if added will snatch Mg^{2+} ions from Mg – indicator complex (because Mg - EDTA complex is more stable than Mg – indicator complex) leaving the bivalent indicator ion free, which being blue in colour will produce blue colour solution. Thus a colour change from red to blue will indicate the end point. The reactions involved are,



Procedure

Titration I Standardization of EDTA

A standard hard water solution is prepared in 100 ml volumetric flask. 20 ml of this solution is pipetted out into a clean conical flask. 2ml of ammonia buffer solution and 2 drops of Eriochrome Black– T indicator are added and the solution turns wine red in colour. Contents of the conical flask are then titrated against EDTA solution taken in the burette. A colour change from wine red to blue is the end point. The titration is repeated to get concordant value.

Titration II Estimation of Total Hardness of water

Given hard water is made upto the mark in 100 ml volumetric flask. 20ml of this solution is pipetted out into a clean conical flask. 2ml of ammonia buffer solution and 2 drops of Eriochrome Black–T indicator are added and solution turns wine red colour. Contents of the conical flask are titrated against standard EDTA solution taken in burette. A colour change from wine red to blue is the end point. Titration is repeated for concordant value. From titre value the total hardness of sample water is determined.

Result

The total hardness present in given water sample = _____ ppm

Titration I**Standardization of EDTA**

Burette Solution	-	EDTA
Pipette Solution	-	20 ml of Zinc Sulphate
Indicator	-	Eriochrome Black– T
End point	-	Colour change from wine red to blue
Additional Solution	-	2 ml of ammonia buffer solution

Standard Zinc sulphate Vs EDTA

Sl. No.	Volume of Zinc sulphate (V ₁) (ml)	Burette Reading (ml)		Volume of EDTA (V ₂) (ml)	Concordant titre value (ml)
		Initial	Final		
1	20				
2	20				
3	20				
4	20				

Volume of zinc sulphate V₁ = 20 ml

Normality of zinc sulphate N₁ = N

Volume of EDTA V₂ = ml (titre value)

Normality of EDTA N₂ = ?

$$V_1 N_1 = V_2 N_2 \quad \quad N_2 = V_1 N_1 / V_2$$

$$\text{Volume of ZnSO}_4 \times \text{Normality of ZnSO}_4$$

$$\text{Normality of EDTA} = \frac{\text{Volume of ZnSO}_4 \times \text{Normality of ZnSO}_4}{\text{Volume of EDTA}}$$

$$N_2 =$$

$$N_2 = \quad \quad N$$

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ESTIMATION OF MAGNESIUM BY EDTA METHOD

Aim

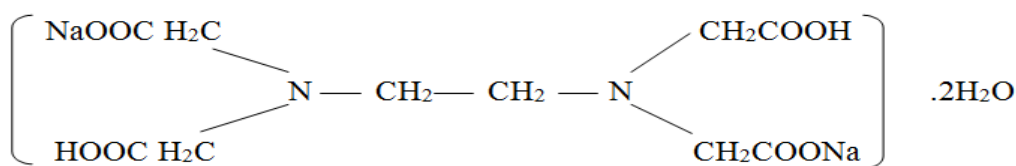
To estimate the amount of magnesium present in the whole of the given solution.

Chemicals Required

EDTA, MgSO_4 , Zinc Sulphate, Ammonia buffer, Eriochrome black – T indicator

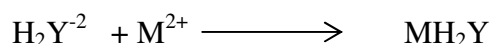
Principle

This method is based upon the fact that EDTA forms stable complexes with metals, particularly with di-and polyvalent metals. Formula of hydrated disodium salt of EDTA

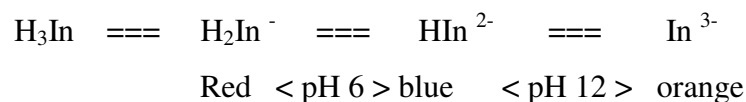


It is abbreviated as $\text{Na}_2\text{H}_2\text{Y}$ and its dissociation $\text{Na}_2\text{H}_2\text{Y} \longrightarrow \text{H}_2\text{Y}^{2-} + 2\text{Na}^+$

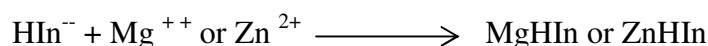
EDTA reacts with divalent metals ions like Zn^{2+} and Mg^{2+} to form stable complexes.



The indicator Eriochrome Black – T is a tribase acid dye represented by H_3In and dissociates as follows producing different coloured ions at various pH values.



The bivalent blue coloured indicator ion which exist at about pH10 reacts with Mg^{2+} or Zn^{2+} producing Mg- Indicator or Zn-Indicator complexes which are wine red in colour.



The metal - EDTA complexes are more stable than metal-indicator complexes. When a solution containing Zn or Mg-ions is titrated with EDTA in the presence of the indicator, the EDTA first reacts with free metals ions to form metal-EDTA complex and at the end point the metal is displaced from the metal-indicator complex by EDTA leaving the free indicator ion which is blue in colour. Therefore the change from wine red to blue is the end point.

Titration II**Estimation of Magnesium sulphate**

Burette Solution	-	EDTA
Pipette Solution	-	20 ml of Magnesium sulphate
Indicator	-	Eriochrome Black– T
End point	-	Colour change from wine red to blue
Additional Solution	-	2 ml of ammonia buffer solution

Standard EDTA Vs Magnesium sulphate

Sl. No.	Volume of MgSO ₄ (V ₁) (ml)	Burette Reading (ml)		Volume of EDTA (V ₂) (ml)	Concordant titre value (ml)
		Initial	Final		
1	20				
2	20				
3	20				
4	20				

Volume of EDTA $V_1 =$ ml

Normality of EDTA $N_1 =$ N

Volume of Magnesium sulphate $V_2 = 20$ ml

Normality of Magnesium sulphate $N_2 = ?$

$$V_1 N_1 = V_2 N_2$$

$$N_2 = V_1 N_1 / V_2$$

Volume of EDTA x Normality of EDTA

Normality of MgSO₄ = _____

Volume of MgSO₄

$$N_1 =$$

Amount of Magnesium = Normality of Magnesium x Eq.wt (12.16) x (100 / 1000)

=

=

g

Procedure

Titration I

Standardization of EDTA

A standard solution of zinc sulphate is prepared in a clean 100 ml SMF. 20 ml of standard zinc sulphate solution is pipetted out in a clean conical flask. 2ml of buffer solution ($\text{NH}_4\text{Cl} + \text{NH}_4\text{OH}$) and 1-2 drops of Eriochrome Black-T indicator are added and the wine red coloured solution is titrated against EDTA taken in the burette until the colour changes to blue. The titration is repeated for concordant values.

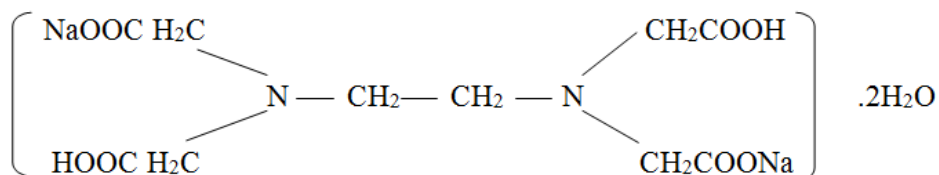
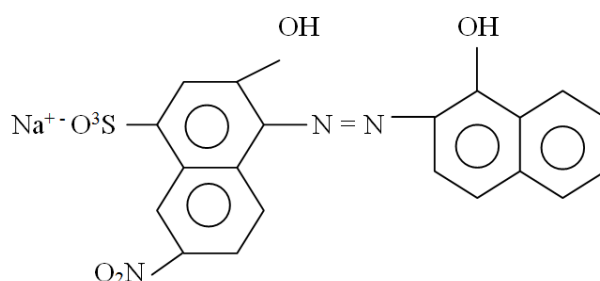
Titration II

Estimation of Magnesium sulphate

The given unknown magnesium sulphate solution is made upto the mark in a clean 100 ml SMF. 20 ml of this solution is pipetted out in a clean conical flask. 2ml of buffer solution ($\text{NH}_4\text{Cl} + \text{NH}_4\text{OH}$) and 1-2 drops of Eriochrome Black-T indicator are added and the wine red coloured solution is titrated against EDTA taken in the burette until the colour changes to blue. The titration is repeated for concordant values.

Result

The amount of magnesium present in the whole of the given solution = g

Viva**1. Write the Structure of EDTA.****Ethylene Diamine Tetra Acetic acid (EDTA)****2. Write the structure of EBT.****3. What is hardness?**

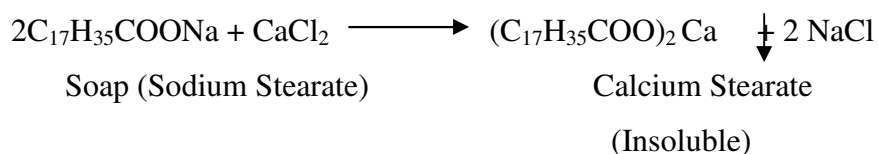
Hardness in water is that characteristic which “prevents lathering of soap”. Originally it was defined as the soap consuming capacity of water sample.

4. Why it is due to hardness?

This is due to presence in water of certain salts of Ca, Mg and other heavy metal ions like Al^{3+} , Fe^{3+} and Mn^{2+} dissolved in it.

5. Why soaps are not effective in hard water?

A sample of hard water, when treated with soap does not produce lather, but on the other hand, forms insoluble white scum (or) precipitates which do not possess any detergent action. This is due to the formation of insoluble soaps of Calcium and Magnesium. Reactions of soap with calcium chloride and magnesium sulphate are,

**6. Explain any two methods of removing permanent hardness.**

- (a) Lime Soda process (b) Zeolite process

7. Difference between temporary and permanent hardness.

Temporary Hardness	Permanent Hardness
The presence of dissolved bicarbonates of Ca^{2+} , Mg^{2+} and other heavy metals and the carbonate of ion.	The presence of dissolved Chloride & sulphates of calcium, Magnesium and other heavy metals.

8. Why metal-EDTA complexes are more stable compared to metal-EBT complexes?

The great stability undoubtedly results from the several complexing sites within the molecule that give rise to a cage like structure in which the cation is effectively surrounded and isolated from solvent molecules.

9. What is the principle of EDTA Titration?

The quick, complete and 1:1 interaction of metal ions with EDTA leading to the formation of stable complex is the basis of complexometric titrations.

10. What are the units of hardness?

- (a) **Parts per million:** It is defined as the parts of calcium carbonate equivalent hardness per 10^6 parts of water.
- (b) **Milligrams per litre :** It is defined as the number of milligrams of calcium carbonate equivalent hardness present per litre of water.

11. What is buffer solution?

The chemical substance which when added to the chemical reaction does not alter the pH of the reaction is known as buffer solution.

12. What are the different types of buffer solution? Give some examples.

1. Acidic buffer, Example : CH_3COOH and CH_3COONa
2. Basic buffer Example : NH_4Cl and NH_4OH

13. Mention the role of buffer solution in a reaction.

It maintains the pH of the solution in a reaction.

Titration I**Standardization of KMnO_4**

Burette Solution	-	KMnO_4
Pipette Solution	-	20 ml of oxalic acid
Indicator	-	Self indicator
End point	-	Appearance of light pink colour

Standard $\text{H}_2\text{C}_2\text{O}_4$ Vs KMnO_4

Sl. No.	Volume of Oxalic acid (V_1) (ml)	Burette Reading (ml)		Volume of KMnO_4 (V_2) (ml)	Concordant titre value (ml)
		Initial	Final		
1	20				
2	20				
3	20				
4	20				

Volume of oxalic acid $V_1 = 20$ ml

Normality of oxalic acid $N_1 =$ N

Volume of KMnO_4 $V_2 =$ ml

Normality of KMnO_4 $N_2 = ?$

$$V_1 N_1 = V_2 N_2 \quad N_2 = V_1 N_1 / V_2$$

$$\text{Normality of } \text{KMnO}_4 = \frac{\text{Volume of } \text{H}_2\text{C}_2\text{O}_4 \times \text{Normality of } \text{H}_2\text{C}_2\text{O}_4}{\text{Volume of } \text{KMnO}_4}$$

$$N_2 =$$

$$N_2 = \quad N$$

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ESTIMATION OF FERROUS BY PERMANGANOMETRY

Aim

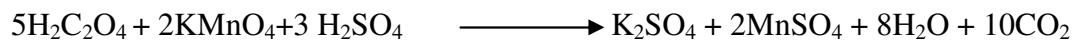
To estimate the amount of ferrous iron in the given solution of ferrous ammonium sulphate.

Chemicals Required

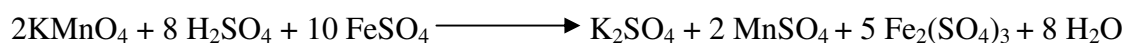
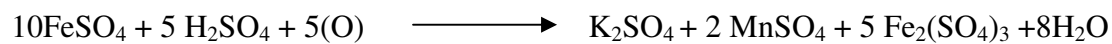
1. KMnO_4
2. Oxalic acid
3. 2N Sulphuric acid
4. Ferrous ammonium sulphate

Principle

KMnO_4 oxidizes oxalic acid to carbon dioxide and water in the presence of H_2SO_4 .



KMnO_4 oxidizes oxalic acid to carbon dioxide and water in the presence of H_2SO_4 .



Titration II**Estimation of Fe^{2+}**

Burette Solution	-	KMnO_4
Pipette Solution	-	20 ml of FAS
Indicator	-	Self Indicator
End point	-	Appearance of light pink colour

Standard KMnO_4 Vs FAS

Sl. No.	Volume of FAS (V_1) (ml)	Burette Reading (ml)		Volume of KMnO_4 (V_2) (ml)	Concordant titre value (ml)
		Initial	Final		
1	20				
2	20				
3	20				

Volume of KMnO_4 $V_1 =$ ml (titre value)

Normality of KMnO_4 $N_1 =$ N

Volume of FAS $V_2 = 20$ ml

Normality of FAS $N_2 = ?$

$$V_1 N_1 = V_2 N_2$$

$$N_2 = V_1 N_1 / V_2$$

Volume of KMnO_4 x Normality of KMnO_4

$$\text{Normality of } \text{Fe}^{2+} = \frac{\text{Volume of } \text{KMnO}_4 \times \text{Normality of } \text{KMnO}_4}{\text{Volume of FAS}}$$

$$N_2 =$$

$$N_2 = \text{N}$$

Amount of $\text{Fe}^{2+} = \text{Normality of FAS} \times \text{Eq.wt (55.85)} \times (100 / 1000)$

=

= g

Procedure

Titration I Standardization of Potassium permanganate

A standard solution of oxalic acid is prepared in a 100 ml volumetric flask. 20 ml of this solution is pipetted out into a clean conical flask and 10 ml of 2N H₂SO₄ is added. The content in the conical flask is heated to 60 – 70° C and the warm solution is titrated against KMnO₄ taken in the burette till the appearance of light pink colour. KMnO₄ acts as a self indicator. The titration is repeated for concordant value.

Titration II Estimation of Ferrous iron

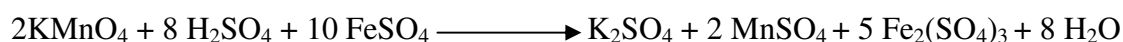
The given solution of ferrous ammonium sulphate is made upto the mark in a clean 100 ml SMF. 20 ml of this solution is pipetted out into a clean conical flask and 10 ml of 2N H₂SO₄ is added. The solution is titrated against KMnO₄ till the appearance of light pink colour. The titration is repeated for concordant values.

Result

The amount of Fe²⁺ present in the whole of the given solution = g

Viva

1. Write the reaction involved in the estimation of ferrous by permanganometry.



2. What is the indicator used in permanganometry?

Permanganate acts as self indicator.

Titration I**Standardization of HCl**

- | | | |
|---------------------|---|---|
| 1. Burette Solution | - | HCl solution |
| 2. Pipette Solution | - | 20 ml of standard Na ₂ CO ₃ |
| 3. Indicator | - | Methyl orange |
| 4. End point | - | Appearance of red colour |

Standard Na₂CO₃ Vs HCl

Sl. No.	Volume of Na ₂ CO ₃ (V ₁) (ml)	Burette Reading (ml)		Volume of HCl solution (V ₂) (ml)	Concordant Titre Value (ml)
		Initial	Final		
1	20				
2	20				
3	20				
4	20				

Volume of standard Na₂CO₃ solution (V₁) = 20 ml

Normality of standard Na₂CO₃ (N₁) = _____ N

Volume of HCl (V₂) = _____ ml (titre value)

Normality of HCl (N₂) = ?

$$V_1 N_1 = V_2 N_2 \quad N_2 = V_1 N_1 / V_2$$

$$\text{Normality of HCl} = \frac{\text{Volume of Na}_2\text{CO}_3 \times \text{Normality of Na}_2\text{CO}_3}{\text{Volume of HCl}}$$

$$N_2 =$$

Normality of HCl (N₂) = _____ N

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Estimation of Mixture of Alkali

Aim

To estimate the amount of alkali (carbonate and bicarbonate) present in the given solution.

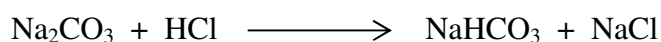
Chemicals Required

1. Standard HCl
2. Na_2CO_3
3. Methyl Orange Indicator
4. Phenolphthalein Indicator

Principle

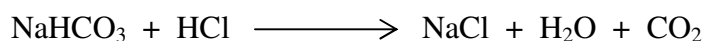
Alkalinity is caused by the presence of hydroxide, carbonate and bicarbonate. The various alkalinities can be estimated by titrating with a standard acid using phenolphthalein and methyl orange indicators successively.

When standard solution of strong acid is added to alkali mixture, half neutralization of carbonate to bicarbonates takes place during phenolphthalein end point.



The phenolphthalein indicator changes from red to colourless at about pH 8.

At this point if methyl orange is added, the bicarbonate neutralization occurs and the pH ultimately drops to about 4.



As the methyl orange changes from yellow to pink at this pH, the occurrence of the complete neutralization can be detected in this step, any bicarbonate present plus the bicarbonate produced in the first step.

If X ml of HCl were required with phenolphthalein indicator and Y ml of the same acid with methyl orange indicator, then 2X ml of acid corresponds to neutralization of carbonate and (Y-2X) ml of acid to bicarbonate neutralization

Titration II**Estimation of mixture of alkali**

1. Burette Solution - HCl acid
2. Pipette Solution - 20 ml of alkali mixture solution
3. Indicator - Phenolphthalein and methyl orange
4. End point - Disappearance of pink colour and then appearance of red orange colour
5. Equivalent weight of Na_2CO_3 - 53
6. Equivalent weight of NaHCO_3 - 84

HCl Vs Mixture of alkali

Sl. No.	Volume of alkali mixture (V_2) (ml)	Burette Readings (ml)			Volume of HCl for Carbonate X (ml)	Volume of HCl for Bicarbonate Y (ml)	Concordant titre value (ml)	
		Initial	Phenolphthalein End point	Methyl Orange End point			X	Y
1	20							
2	20							
3	20							
4	20							

STEP I

Volume of HCl for Phenolphthalein end point (X) = _____ ml

Volume of HCl for Carbonate (V_1) = $2X$ = _____ ml

Normality of HCl solution (N_1) = _____ N

Volume of mixture of alkali solution (V_2) = 20 ml

Normality of Carbonate in the mixture (N_2) = $(V_1 \times N_1) / V_2$

Normality of HCl = { Volume of HCl x Normality of HCl } / Volume of Alkali mixture
=

Normality of Carbonate in the mixture = _____ N

Amount of Carbonate present = Normality of Carbonate x Eq.Wt (53) x 1000

=

= _____ mg / litre

Procedure

Titration I Standardization of HCl solution

A standard solution of Sodium carbonate is prepared in a 100 ml volumetric flask. 20 ml of this solution is pipetted out in a clean conical flask. Two drops of methyl orange indicator is added and titrated against HCl taken in the burette. The colour change from yellow to red colour is the end point. Titrations are repeated to get concordant value. From the value of HCl consumed strength of HCl is calculated.

Titration II Estimation of Carbonate and Bicarbonate

STEP I (with Phenolphthalein indicator)

The given alkali mixture is made upto the mark in a 100 ml volumetric flask. 20 ml of this solution is taken into a clean conical flask and 2 drops of phenolphthalein indicator is added and the solution turns pink in colour. This solution is titrated against HCl from the burette until the pink colour disappears. The volume of acid thus required is equivalent to half neutralization of carbonate alkalinity.

STEP II (with Methyl orange indicator)

2 drops of methyl orange indicator is now added to the above colourless solution and the solution turns yellow in colour. The titration is then continued with the same HCl till the appearance of red orange colour, indicating the complete neutralization of bicarbonates. The volume of acid used for this is equivalent to any bicarbonate already present plus the bicarbonate produced by the half neutralization of carbonate in the first step.

The titrations are repeated to get concordant value. From the known values of HCl consumed in the titrations, the strength of carbonate and bicarbonate are calculated. From this their amounts are calculated.

Result

Amount of carbonate present in whole of the given solution = _____ mg / litre.

Amount of bicarbonate present in whole of the given solution = _____ mg / litre.

STEP II

Volume of HCl required for total
neutralization of $\text{Na}_2\text{CO}_3 + \text{NaHCO}_3$ = _____ (Y) ml

Volume of HCl that corresponds to
complete neutralization of $\text{NaHCO}_3 = (Y - 2X)$ = _____ (Z) ml

Normality of HCl solution (N_1) = _____ N

Volume of mixture of alkali solution (V_2) = 20 ml

Normality of Bicarbonate in the mixture (N_2) = $Z \times N_1 / V_2$

$$\text{Normality of Bicarbonate} = \frac{\text{Volume of HCl} \times \text{Normality of HCl}}{\text{Volume of alkali mixture}}$$

=

Normality of Bicarbonate = _____ N

Amount of Bicarbonate present = Normality of Bicarbonate x Eq.Wt (84) x 1000

=

Amount of Bicarbonate = _____ mg / litre

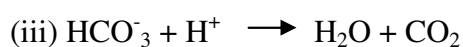
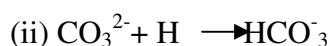
1. What is Alkalinity?

Alkalinity is caused by the presence of hydroxide, carbonate and bicarbonate. The various alkalinities can be estimated by titrating with a standard acid using phenolphthalein and methyl orange indicators successively it is known as alkalinity.

2. How can you estimate the alkalinity?

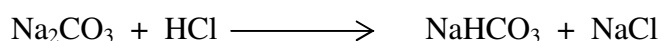
These can be estimated separately by titration against standard acid, using phenolphthalein and methyl orange as indicators.

The determination is based on the following reactions:



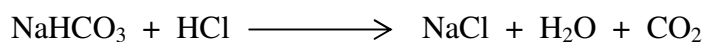
3. Write the Reaction and Equation , involved in alkali mixture?

When a standard solution of strong acid is added to the alkali mixture, the half neutralization of carbonate to bicarbonates takes place during phenolphthalein end point.



The phenolphthalein indicator changes from red to colourless at about pH 8.

At this point if methyl orange is added, the bicarbonate neutralization occurs and the pH ultimately drops to about 4.



As the methyl orange changes from yellow to pink at this pH, the occurrence of the complete neutralization can be detected in this step, any bicarbonate present plus the bicarbonate produced in the first step.

4. What is the equivalent weight of Na_2CO_3 & NaHCO_3 ?

Equivalent weight of Na_2CO_3 - 53

Equivalent weight of NaHCO_3 - 84.

5. What are Neutralization Reactions?

Acid –Base combined to form salt and water is known as Neutralization Reactions

Example : $\text{HCl} + \text{NaOH} \longrightarrow \text{NaCl} + \text{H}_2\text{O}$

Titration I**Standardization of Sodium thiosulphate**

Burette Solution	-	Sodium thiosulphate
Pipette Solution	-	20 ml of standard potassium dichromate
Indicator	-	1 ml of 1% Starch solution
End point	-	Appearance of pale green colour
Additional Solution	-	10 ml of 2N HCl + 5 ml of 10% KI

Standard $K_2Cr_2O_7$ Vs $Na_2S_2O_3$

Sl. No.	Volume of $K_2Cr_2O_7$ solution (V_1) (ml)	Burette Readings (ml)		Volume of $Na_2S_2O_3$ solution (V_2) (ml)	Concordant titre value (ml)
		Initial	Final		
1	20				
2	20				
3	20				
4	20				

Volume of Potassium dichromate solution (V_1) = 20 ml

Normality of Potassium dichromate solution (N_1) = _____ N

Volume of Sodium thiosulphate solution (V_2) = _____ ml

Normality of Sodium thiosulphate solution (N_2) = ?

$$V_1 N_1 = V_2 N_2 \quad N_2 = V_1 N_1 / V_2$$

Volume of Dichromate x Normality of Dichromate

Normality of Thio = $\frac{\text{Volume of Dichromate} \times \text{Normality of Dichromate}}{\text{Volume of Thio}}$

$$N_2 =$$

Normality of sodium thiosulphate = _____ N

Expt. No:

Date:

DETERMINATION OF DISSOLVED OXYGEN IN WATER BY WINKLER'S METHOD

Aim

To determine dissolved oxygen present in water sample by Winkler's method.

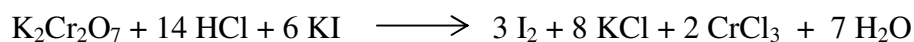
Chemicals Required

Standard $K_2Cr_2O_7$ solution	Sodium thiosulphate
40% Manganous sulphate	Alkaline KI solution
1% Starch indicator	Conc. H_2SO_4

Principle

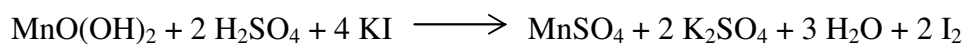
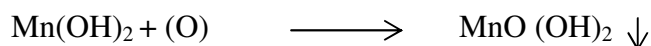
Standardization of Sodium thiosulphate

Potassium dichromate oxidises an acidified solution of KI to an equivalent amount of iodine. Iodine liberated is titrated against $Na_2S_2O_3$ using starch indicator.

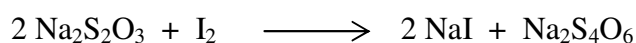


Estimation of dissolved oxygen

Oxygen dissolves in water to the extent of 7 – 9 mg/litre at a temperature range of 25° – 35° C. The estimation of dissolved oxygen is based on the fact that atomic oxygen decomposes potassium iodide liberating free iodine. This can be titrated against a standard solution of sodium thiosulphate using starch indicator. But oxygen present in water is in molecular form, its conversion into atomic state is necessary. This is done by reacting the water sample with manganous hydroxide formed by the reaction of manganous sulphate with potassium hydroxide.



This liberated iodine is titrated with standard sodium thiosulphate solution.



Titration II**Estimation of Dissolved Oxygen in water sample**

Burette Solution	-	Sodium Thiosulphate solution
Pipette Solution	-	100 ml of Water sample
Indicator	-	1 ml of 1% Starch solution
End point	-	Disappearance of blue colour
Additional Solution	-	2ml MnSO ₄ + 2 ml Alkaline KI + Conc. H ₂ SO ₄
Eq. Wt. of Oxygen	-	8

Sodium Thiosulphate Vs Water sample

Sl. No.	Volume of Water sample (V ₁) (ml)	Burette Readings (ml)		Volume of Sodium Thiosulphate (V ₂) (ml)	Concordant titre value (ml)
		Initial	Final		
1	50				
2	50				

Volume of Sodium Thiosulphate (V₁) = _____ ml

Normality of Sodium Thiosulphate (N₁) = _____ N

Volume of Water sample (V₂) = 50 ml

Normality of Dissolved Oxygen (N₂) = ?

$$V_1 N_1 = V_2 N_2 \quad N_2 = V_1 N_1 / V_2$$

Normality of Dissolved Oxygen = { Volume of Sodium Thiosulphate x Normality of Sodium Thiosulphate } / Volume of Water sample

$$N_2 =$$

Normality of Dissolved Oxygen (N₂) = _____ N

Amount of Dissolved Oxygen present in one litre of water sample

$$= \{ \text{Normality of Dissolved oxygen} \times \text{Eq. Wt of Oxygen (8)} \times 1000 \}$$

$$= \{ \text{_____ N} \times 8 \times 1000 \}$$

Amount of Dissolved Oxygen = _____ ppm

Procedure

Titration I Standardization of Sodium thiosulphate

A standard solution of potassium dichromate is prepared in a 100 ml volumetric flask. 20 ml of potassium dichromate solution is pipetted out into a clean conical flask. About 10 ml of 2N HCl and 5 ml of 10% KI are added. The liberated iodine is titrated against sodium thiosulphate solution taken in the burette. When the iodine solution becomes straw yellow in colour, 1 ml of freshly prepared starch indicator is added and the titration is continued carefully. At the end point, blue colour will suddenly discharge and a pale green colour takes place due to the chromic salt. The titrations are repeated to get concordant value. From the titre value, the strength of sodium thiosulphate is calculated.

Titration II Estimation of Dissolved Oxygen

The given water sample is taken in a 250 ml reagent bottle leaving 10 ml space. To this, 2ml of manganous sulphates solution and 2ml of alkaline KI solution are added and precipitate of manganous hydroxide will be formed. The bottle is shaken well to facilitate the reaction of Mn(OH)_2 precipitate with the dissolved oxygen present in the water sample. When all the dissolved oxygen in water has reacted with Mn(OH)_2 , it gives a brown precipitate of basic manganic hydroxide. Now $\text{Conc.H}_2\text{SO}_4$ is added drop by drop to dissolve the precipitate till yellow solution of iodine is obtained.

100ml of this iodine solution is taken into a clean conical flask and titrated against sodium thiosulphate solution using starch indicator. Disappearance of blue colour is the end point. From the titre value the strength of dissolved oxygen is calculated and from this the amount of dissolved oxygen in the water sample is calculated.

Result

The amount of Dissolved Oxygen in the given water sample = _____ ppm

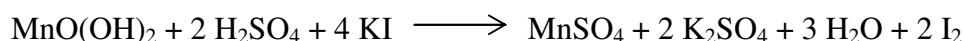
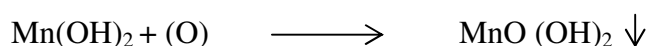
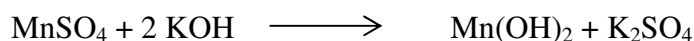
Viva

1. What is the amount of oxygen dissolved in water at room temperature?

Oxygen dissolves in water to the extent of 7 – 9 mg/litre at a temperature range of 25° – 35° C.

2. Write the reaction involved in the estimation of dissolved oxygen.

The water sample reacts with manganous hydroxide formed by the reaction of manganous sulphate with potassium hydroxide.



This liberated iodine is titrated with standard sodium thiosulphate solution.

3. Write down the formula used to calculate the dissolved oxygen. What is the unit of dissolved oxygen?

Dissolved Oxygen = Normality of Dissolved oxygen X Equivalent weight of Oxygen (8) X 1000, Unit of dissolved oxygen = ppm (parts per million)

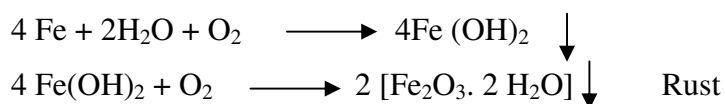
4. Discuss the factors which affect the solubility of oxygen in water.

- (a) Temperature: Solubility of oxygen in water decreases with increase in temperature
- (b) Concentration of impurities: Solubility of dissolved oxygen in water decreases with increase in the concentration of impurities.

5. Explain why the determination of dissolved oxygen is important for industrial purpose. Give an example.

Dissolved oxygen is the most usual corrosion causing factor in boilers. Oxygen is introduced in boilers through the raw water supply. Water usually contains about 8 ppm of dissolved oxygen at room temperature. As the water is heated, the dissolved oxygen is set free and the boiler starts corroding.

Dissolved oxygen reacts with the iron of boiler in presence of water and under prevailing high temperature to form ferric oxide rust.



6. Mention some examples for iodometry estimations.

- (i) Determination of Dissolved Oxygen in water.(ii) Estimation of available Chlorine in Bleaching powder.iii) Determination of Copper in Brass

Titration I**Standardization of Thiosulphate**

Burette Solution	-	Sodium Thiosulphate
Pipette Solution	-	20 ml of standard potassium dichromate
Indicator	-	1 ml of 1% Starch solution
End point	-	Appearance of pale green Colour
Additional Solution	-	10 ml of 2N HCl + 5 ml of 10% KI

Standard $K_2Cr_2O_7$ Vs $Na_2S_2O_3$

Sl. No.	Volume of $K_2Cr_2O_7$ solution (V_1) (ml)	Burette Readings (ml)		Volume of $Na_2S_2O_3$ solution (V_2) (ml)	Concordant titre value (ml)
		Initial	Final		
1	20				
2	20				
3	20				

Volume of Potassium dichromate solution (V_1) = 20 ml

Normality of Potassium dichromate solution (N_1) = _____ N

Volume of Sodium thiosulphate solution (V_2) = _____ ml

Normality of Sodium thiosulphate solution (N_2) = ?

$$V_1 N_1 = V_2 N_2 \quad N_2 = V_1 N_1 / V_2$$

Volume of Dichromate x Normality of Dichromate

Normality of Thio = _____
Volume of Thio

$$N_2 =$$

Normality of sodium thiosulphate = _____ N

Expt. No:

Date:

ESTIMATION OF AVAILABLE CHLORINE IN BLEACHING POWDER

Aim

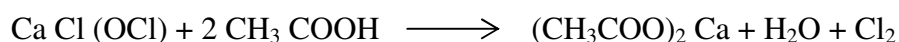
To estimate the amount of available chlorine in given bleaching powder solution.

Chemicals Required

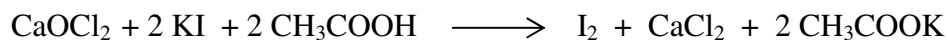
Potassium dichromate	Sodium thiosulphate
Bleaching powder	2N HCl
2N Acetic acid	10% KI
Starch indicator	

Principle

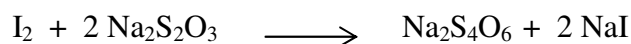
The commercial sample of bleaching powder is a mixture of calcium hydroxide, calcium chloride, calcium hypochlorite $\text{Ca}(\text{OCl}_2)$ and calcium chlorate $\text{Ca}(\text{ClO}_3)_2$. When bleaching powder is treated with dilute acid, chlorine is evolved. The chlorine liberated is known as available chlorine. Estimation of available chlorine is useful in determining the quality of bleaching powder. Commercial sample has 36 – 38% of available chlorine.



A known volume of a solution of bleaching powder is treated with excess of potassium iodide and acidified with acetic acid.



Iodine liberated is titrated against sodium thiosulphate solution using starch indicator.



From the amount of liberated iodine, the equivalent of Cl_2 and hence the available chlorine in the sample is calculated.

Titration II**Estimation of Available chlorine**

Burette Solution	-	Sodium Thiosulphate solution
Pipette Solution	-	20 ml of bleaching powder solution
Indicator	-	1 ml of 1% Starch solution
End point	-	Disappearance of blue colour
Additional Solution	-	20 ml dil. Acetic acid + 5 ml of 10% KI
Eq. Wt. of Chlorine	-	35.45

Sodium Thiosulphate Vs Bleaching powder solution

Sl. No.	Volume of Bleaching powder solution (V ₁) (ml)	Burette Reading (ml)		Volume of Sodium Thiosulphate (V ₂) (ml)	Concordant titre value (ml)
		Initial	Final		
1	20				
2	20				
3	20				

Volume of Sodium Thiosulphate solution (V₁) = _____ ml

Normality of Sodium Thiosulphate solution (N₁) = _____ N

Volume of Bleaching powder solution (V₂) = 20 ml

Normality of Chlorine (N₂) = ?

$$V_1 N_1 = V_2 N_2 \quad N_2 = V_1 N_1 / V_2$$

Normality of Chlorine = { Volume of Sodium Thiosulphate x Normality of Sodium Thiosulphate } / Volume of Bleaching powder

$$N_2 =$$

Normality of Chlorine (N₂) = _____ N

Amount of chlorine = Normality of chlorine x Eq. Wt of chlorine (35.45) x (100/1000)

$$= \text{_____ N} \times 35.45 \times (100 / 1000)$$

Amount of chlorine present = _____ g

Procedure

Titration I Standardization of sodium thiosulphate

Prepare a standard solution of potassium dichromate in a 100 ml volumetric flask. Pipette out 20 ml of potassium dichromate into a clean conical flask. Add about 10 ml of 2N HCl and 5 ml of 10% KI. Titrate the liberated iodine against sodium thiosulphate solution taken in the burette. When the iodine solution becomes pale yellow in colour add 1 ml of freshly prepared starch solution and carefully continue the titration. At the end point blue colour will suddenly discharge and a pale green colour takes place due to the chromic salt. Repeat the titrations for concordant value. From the titre value strength of sodium thiosulphate is calculated.

Titration II Estimation of available Chlorine

Make up the given solution of bleaching powder up to the mark in 100 ml volumetric flask. Pipette out 20 ml of this solution into a clean conical flask. Add about 5 ml of 10% KI solution and 10 ml of 2N acetic acid. The solution turns dark brown in colour due to the liberated iodine. Titrate the liberated iodine against sodium thiosulphate. When the solution becomes straw yellow, 1 ml of freshly prepared starch indicator is added and the titration is continued. End point is the disappearance of blue colour. Repeat the titration for concordant values. From the titre value the chlorine liberated and hence its amount in the sample is calculated.

Result

Amount of available chlorine present in whole of the given sample of bleaching powder solution = _____ g

Viva**1. What is meant by Dichrometry?**

A standard solution of Potassium dichromate is titrated against acidified solution of a ferrous salt. Potassium dichromate acts as a good oxidizing agent in presence of acids.

**2. What is meant by Iodometry?**

The liberated Iodine(KI) can be titrated against the standard sodium thiosulphate solution in presence of oxidizing agent.

3. What is meant by Complexometric titrations?

Estimation of the solution involving complex formation by using a complexing agent EDTA and a complexing indicator Eriochrome Black-T

4. Write down the ingredient of commercial bleaching powder.

The commercial sample of bleaching powder is a mixture of,

Calcium hydroxide $[\text{Ca}(\text{OH}_2)]$, Calcium chloride $[\text{CaCl}_2]$

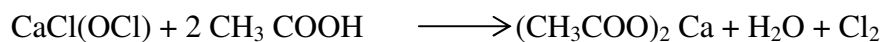
Calcium hypochlorite $[\text{Ca}(\text{OCl}_2)]$ Calcium chlorate $[\text{Ca}(\text{ClO}_3)_2]$

5. What is meant by available chlorine in bleaching powder?

When bleaching powder is treated with dilute acid, chlorine is evolved. The chlorine liberated is known as available chlorine. Estimation of available chlorine is useful in determining the quality of bleaching powder.

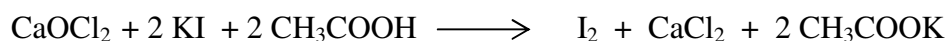
6. Write the reactions in estimation available chlorine in bleaching powder.

Estimation of available chlorine is useful in determining the quality of bleaching powder. Commercial sample has 36 – 38% of available chlorine.



Bleaching powder

A known volume of a solution of bleaching powder is treated with excess of potassium iodide and acidified with acetic acid.



The iodine liberated is ~~titrated~~ ^{titrated} against sodium thiosulphate solution using starch indicator. $\text{I}_2 + 2 \text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{Na}_2\text{S}_4\text{O}_6 + 2 \text{NaI}$

From the amount of liberated iodine, the equivalent of Cl_2 and hence the available chlorine in the sample is calculated

Titration I**Standardization of Sodium thiosulphate**

Burette Solution	-	Sodium Thiosulphate
Pipette Solution	-	20 ml of standard Potassium dichromate
Indicator	-	Starch solution
End point	-	Appearance of pale green colour
Additional Solution	-	10 ml of 2N HCl + 5 ml of 10% KI

Standard $K_2Cr_2O_7$ Vs $Na_2S_2O_3$

Sl. No.	Volume of $K_2Cr_2O_7$ solution (V_1) (ml)	Burette Readings (ml)		Volume of $Na_2S_2O_3$ solution (V_2) (ml)	Concordant titre value (ml)
		Initial	Final		
1	20				
2	20				
3	20				
4	20				

Volume of Potassium dichromate solution (V_1) = 20 ml

Normality of Potassium dichromate solution (N_1) = _____ N

Volume of Sodium thiosulphate solution (V_2) = _____ ml (titre value)

Normality of Sodium thiosulphate solution (N_2) = ?

$$V_1 N_1 = V_2 N_2 \quad N_2 = V_1 N_1 / V_2$$

Volume of Dichromate x Normality of Dichromate

Normality of Thio = _____
Volume of Thio

$$N_2 =$$

Normality of sodium thiosulphate = _____ N

Expt. No:

Date:

ESTIMATION OF COPPER

Aim

To estimate the amount of copper present in given solution.

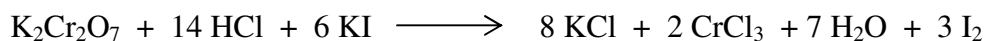
Chemicals Required

Potassium dichromate	Sodium Thiosulphate
Copper sulphate	Conc. HNO ₃
10 % KI	Starch indicator
2N HCl	

Principle

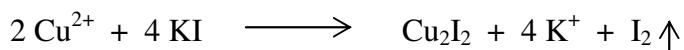
Standardization of Sodium thiosulphate

Potassium dichromate oxidizes an acidified solution of KI to an equivalent of Iodine. The iodine liberated is titrated against Sodium thiosulphate using starch as indicator.

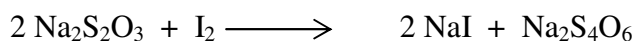


Estimation of Copper

The determination of Copper is based on iodometric titration. A known quantity of Copper ore is dissolved in Conc. HNO₃ and the nitrate is removed by fuming with Conc. H₂SO₄ and adjusting the pH by NH₄OH. Copper reacts with KI in acetic acid medium. Copper is precipitated as cuprous iodide and an equivalent amount of iodine is liberated.



The liberated iodine is titrated against standard sodium thiosulphate using starch indicator.



Titration II**Estimation of Copper**

Burette Solution	-	Sodium Thiosulphate
Pipette Solution	-	20 ml of Copper solution
Indicator	-	Starch solution
End point	-	Appearance of dirty white colour
Additional solution	-	Acetic acid + NH_3 + 5 ml of 10% KI
Eq. Wt. of Copper	-	63.5

Sodium Thiosulphate Vs Copper solution

Sl. No.	Volume of Copper solution (V_1) (ml)	Burette Readings (ml)		Volume of Sodium Thiosulphate (V_2) (ml)	Concordant titre value (ml)
		Initial	Final		
1	20				
2	20				
3	20				
4	20				

Volume of Sodium Thiosulphate solution (V_1) = _____ ml (titre value)

Normality of Sodium Thiosulphate solution (N_1) = _____ N

Volume of Copper solution (V_2) = 20 ml

Normality of Copper solution (N_2) = ?

$$V_1 N_1 = V_2 N_2 \quad N_2 = V_1 N_1 / V_2$$

Normality of Copper = { Volume of Thio x Normality of Thio } / Volume of Copper

$$N_2 =$$

Normality of Copper solution (N_2) = _____ N

Amount of Copper = Normality of Copper solution x Eq. Wt. of Copper x (100/1000)

$$= \quad \quad \quad \text{N} \times 63.5 \times (100 / 1000)$$

Amount of Copper = _____ g

Procedure

Titration I

Standardization of sodium thiosulphate

A standard solution of potassium dichromate is prepared in a 100 ml volumetric flask. 20 ml of potassium dichromate is pipette out into a clean conical flask. About 10 ml of 2N HCl and 5 ml of 10% KI are added and the liberated iodine is titrated against sodium thiosulphate solution taken in the burette. When the iodine solution becomes pale yellow in colour, 1 ml of freshly prepared starch solution is added and the titration is continued. At the end point, blue colour will suddenly discharge and a pale green colour takes place due to the chromic salt. The titrations are repeated for concordant value. From the titre value, the strength of sodium thiosulphate is calculated.

Titration II

Estimation of Copper

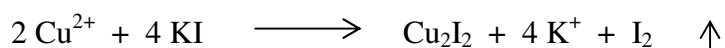
The given copper solution is made upto the mark in a 100 ml volumetric flask. 20 ml of the copper solution is pipetted out into a clean conical flask and 1:1 NH_3 is added drop by drop till a faint blue colour appears (cupric hydroxide). The alkali destroys the free mineral acid. Glacial acetic acid is added in drops to make the solution clear. About 5 ml of 10% KI is added and the liberated iodine is titrated against the sodium thiosulphate. When the solution becomes pale yellow colour, 1 ml of freshly prepared starch indicator is added and the titration is continued. At the end point blue colour suddenly changes to dirty white due to the precipitate of cuprous iodide. The titrations are repeated for concordant value. From the titre value the amount of copper present in the sample is calculated.

Result

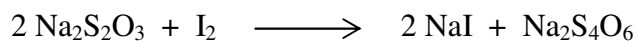
Amount of copper present in whole of the given solution = _____ g

Viva**1. Write the Reaction involved in the estimation of Copper.**

The determination of Copper is based on iodometric titration. A known quantity of Copper ore is dissolved in Conc. HNO_3 and the nitrate is removed by fuming with Conc. H_2SO_4 and adjusting the pH by NH_4OH . Copper reacts with KI in acetic acid medium. Copper is precipitated as cuprous iodide and an equivalent amount of iodine is liberated.



The liberated iodine is titrated against standard sodium thiosulphate using starch indicator.

**2. What is the colour change at the end point in copper estimation? Why the colour changes suddenly?**

The end point is blue colour will suddenly discharge and a pale green colour takes place due to the chromic salt. At the end point blue colour suddenly changes to dirty white due to the precipitate of cuprous iodide.

Titration I Standardization of NaOH

Burette Solution	-	NaOH
Pipette Solution	-	20 ml of oxalic acid
Indicator	-	Phenolphthalein
End point	-	Appearance of pale pink colour

Standard $\text{H}_2\text{C}_2\text{O}_4$ Vs NaOH

Sl. No.	Volume of Oxalic acid (V_1) (ml)	Burette Reading (ml)		Volume of NaOH (V_2) (ml)	Concordant titre value (ml)
		Initial	Final		
1	20				
2	20				
3	20				
4	20				

Volume of oxalic acid $V_1 = 20$ ml

Normality of oxalic acid $N_1 =$ N

Volume of NaOH $V_2 =$ ml

Normality of NaOH $N_2 = ?$

$$V_1 N_1 = V_2 N_2$$

$$N_2 = V_1 N_1 / V_2$$

Volume of $\text{H}_2\text{C}_2\text{O}_4$ x Normality of $\text{H}_2\text{C}_2\text{O}_4$

Normality of NaOH = $\frac{\text{Volume of } \text{H}_2\text{C}_2\text{O}_4 \times \text{Normality of } \text{H}_2\text{C}_2\text{O}_4}{\text{Volume of NaOH}}$

$$N_2 =$$

Normality of NaOH (N_2) = N

Expt. No:

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ESTIMATION OF ACETIC ACID

Aim

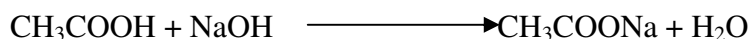
To estimate the amount of acetic acid present in the given unknown solution.

Chemicals Required

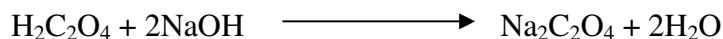
1. Oxalic acid
2. NaOH
3. Phenolphthalein
4. Acetic acid

Principle

A Known weight of acetic acid is diluted with water and the amount of acetic acid is found by titrating it against standard sodium hydroxide solution using phenolphthalein as indicator.



Oxalic acid reacts with sodium hydroxide to give sodium oxalate and water and hence used as primary standard to standardize sodium hydroxide



Procedure:

Titration I

Standardization of NaOH

A standard solution of oxalic acid is prepared in a clean 100 ml standard flask. 20 ml of oxalic acid is pipetted out in a clean conical flask and one drop of phenolphthalein indicator is added. The colourless solution is titrated against NaOH from the burette. The end point is the appearance of pale pink colour. The titration is repeated for concordant values.

Titration II Estimation of Acetic acid

Burette Solution	-	NaOH
Pipette Solution	-	20 ml of Acetic acid
Indicator	-	Phenolphthalein
End point	-	Appearance of pale pink colour

Standard NaOH Vs Acetic acid

Sl. No.	Volume of Acetic acid (V ₁) (ml)	Burette Reading (ml)		Volume of NaOH (V ₂) (ml)	Concordant titre value (ml)
		Initial	Final		
1	20				
2	20				
3	20				
4	20				

Volume of NaOH V₁ = ml

Normality of NaOH N₁ = N

Volume of Acetic acid V₂ = 20 ml

Normality of Acetic acid N₂ = ?

$$V_1 N_1 = V_2 N_2 \qquad \qquad \qquad N_2 = V_1 N_1 / V_2$$

Volume of NaOH x Normality of NaOH

Normality of Acetic acid = $\frac{\text{Volume of NaOH} \times \text{Normality of NaOH}}{\text{Volume of Acetic acid}}$

$$N_2 =$$

Normality of Acetic acid (N₂) = N

Amount of acetic acid = Normality of acetic acid x Eq.wt (60) x (100 / 1000)

$$= \text{N} \times \text{Eq.wt (60)} \times (100 / 1000)$$

Amount of acetic acid = g

Titration II Estimation of Acetic acid

By using a weighing bottle an exactly known weight of the given acetic acid sample is transferred into a clean 100 ml SMF and made up the mark. 20 ml of this solution is pipette out into a clean conical flask and 1 drop of phenolphthalein indicator is added. The solution is titrated against NaOH from the burette. The end point is the appearance of pale pink colour. The titration is repeated for concordant values.

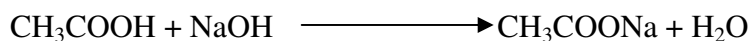
Result

The amount of acetic acid present in the given sample = g

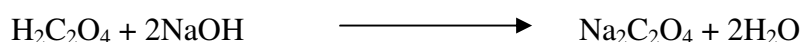
Viva

1. How vinegar is estimated?

A Known weight of vinegar is diluted with water and the percentage of acetic acid is found by titrating it against standard sodium hydroxide solution using phenolphthalein as indicator



Oxalic acid reacts with sodium hydroxide to give sodium oxalate and water and hence used as primary standard to standardize sodium hydroxide



2. Mention the indicator used in the estimation of vinegar?

Phenolphthalein.

